General Manager Dashboard – Conversion Funnel Analytics

1. Project Overview

The **General Manager (GM) Dashboard** is a high-performance analytics platform designed to provide actionable visibility into the customer acquisition lifecycle. In a modern business environment, leads are generated across various channels, but the true challenge for management is understanding **velocity** and **attrition** (drop-off).

This system transforms raw, event-based data from a **NoSQL (MongoDB)** environment into an interactive, visual narrative. By using **Django** for robust data processing and **D3.js** for proportional SVG visualization, the dashboard allows stakeholders to identify exactly where the "leak" in the revenue bucket is occurring at a glance.

Funnel stages:

Lead → Qualified → Approved → Converted

The dashboard helps management teams:

- Monitor lead performance across stages

- Analyze conversion rates between stages

- Identify where leads drop off

- Track overall conversion efficiency

2. Technology Stack

1. Backend:
   - Python
   - Django
   - MongoDB
2. Frontend:
   - HTML
   - JavaScript
   - Tailwind CSS
3.Visualization
   - D3.js

3. Project Structure

```
Funnel_dashboard/
backend/
 ├── manage.py
 ├── Core/
 │   ├── settings.py
 │   ├── urls.py
 │   └── wsgi.py
 │
 ├── dashboard/
 │   ├── models.py
 │   ├── views.py
```

```
│   ├── services.py
│   └── apps.py
frontend/
 ├── dashboard.html
 ├── funnel_viz.js
 └── styles/
 │──seed.py
```

4. Project Setup and Installation

1. Clone Repository:

   git clone <https://github.com/albin680/Albin_Joy-Projects/funnel_dashboard>

   cd funnel_dashboard

2. Install Dependencies:

   pip install django pymongo django-cors-headers

3. Run Django Server:

   python manage.py runserver

   Server will start at: http://127.0.0.1:8000

5. Database Setup

The system uses MongoDB to store lead progression records.

Example document:

```
{
 "lead_id": "L001",
 "stage": "Qualified",
 "timestamp": "2026-03-08T10:00:00",
 "source": "Website",
 "created_date": "2026-03-08"
}
```

Database: funnel_db

Collection: leads

6. Architecture Overview

Data Flow:

> MongoDB → Django Backend API → Frontend Dashboard → D3.js Funnel Visualization

Explanation:

1. Lead data is stored in MongoDB.
2. Django APIs aggregate funnel data.
3. The frontend dashboard fetches API data.
4. D3.js renders the interactive funnel visualization.

7. API Endpoints

GET /api/funnel_dashboard/

Returns funnel metrics including:

- Stage names
- Lead counts
- Conversion percentages
- Overall conversion rate

Example response:


```
{
 "stages":[
  {"stage":"Lead","count":120},
  {"stage":"Qualified","count":80},
  {"stage":"Approved","count":50},
  {"stage":"Converted","count":30}
 ],
 "overall_conversion_rate":25
}
```

8. Conversion Rate Calculation

Example:

Lead = 120

Qualified = 80

Approved = 50

Converted = 30

Lead → Qualified

(80 / 120) × 100 = 66.7%

Qualified → Approved

(50 / 80) × 100 = 62.5%

Approved → Converted

(30 / 50) × 100 = 60%

Overall Conversion Rate

Converted / Lead × 100

30 / 120 × 100 = 25%

9. Funnel Visualization

The funnel visualization is built using D3.js.

Features:

- Funnel stages represented visually
- Width proportional to lead count
- Stage labels and counts displayed
- Conversion percentages between stages
- Tooltip interaction on hover
- Dynamic updates when filters change

10.Error Handling

The system ensures:

- API responses are validated before rendering
- Missing data does not break visualization
- Division by zero errors are prevented
- Numbers are consistently formatted